Migrating from ST25R39xx to ST25R200 multipurpose NFC transceivers

Introduction

The ST25R200 multipurpose NFC transceiver delivers high-end performance in a small 4 × 4 mm package. It brings the convenience of contactless interaction and features to a variety of end applications.

Specifically designed for consumer and industrial end products, this transceiver is a versatile solution for enabling NFC use cases. With its compact size and advanced features, the ST25R200 is an ideal choice for manufacturers looking to incorporate NFC technology in their products.

This migration guide provides instructions how to migrate an existing software project based on ST25R39xx to ST25R200.

The purpose of this application note is to provide the reader with a list of considerations and requirements to ensure a successful migration from ST25R39xx to ST25R200.

This application note applies to the part numbers detailed in the following table:

Table 1. **Applicable products**

Product type	Part number
ST25R39xx	ST253911B ST25R3916 ST253916B
ST25Rx00	ST25R100 ST25R200

For further information on ST25R39xx and ST25R200, refer to the documents listed in the table below.

Related documentation

- DS11793: ST25R3911B datasheet
- DS13541: ST25R3916B datasheet
- DS13658: ST25R200 datasheet
- AN5984: *Layout recommendations for the design of boards with ST25R200*
- UM2890: *RF/NFC abstraction layer (RFAL)*
- AN6065: *Antenna design for ST25R200 device*

1 Terminology

The table below defines acronyms and other terms that are used in this application note.

Table 2. Acronyms

Acronym	Definition
AAT	Automatic antenna tuning
ADC	Analog-to-digital converter
CSO	Capacitance sense output
CSI	Capacitance sense input
IRQ	Interrupt request
MCU	Microcontroller unit
P2P	Peer to peer
ACM	Active communication mode
PCM	Passive communication mode
PCB	Printed circuit board
RC	Resistive capacitive
RF	Radio frequency
RFAL	RF abstraction layer
HAL	Hardware abstraction layer
WU	Wake-up
MRT	Mask receive timer
NRT	No response timer
GPT	General purpose timer
WUT	Wake-up timer

2 Signal and pinout comparison

ST25R39xx and ST25R200 devices are all available in QFN packages.

While ST25R39xx devices come in a 32-pin package (5 x 5 mm), ST25R200 devices have a smaller 24 pin package (4 x 4 mm).

The supply system on ST25R200 is similar to that of ST25R3916B, with the omission of V_{DD_TX}.

The ST25R200 pinout is not pin-to-pin compatible with the other devices.

2.1 Signal layout comparison

The images below contain pinout information for signal layout comparison between ST25R39xx and ST25R200.

Table 3. Comparison between ST25R3911B and ST25R200

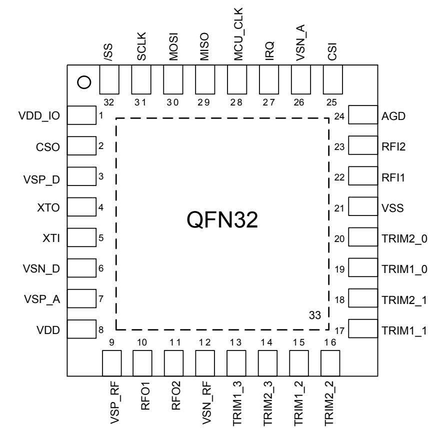
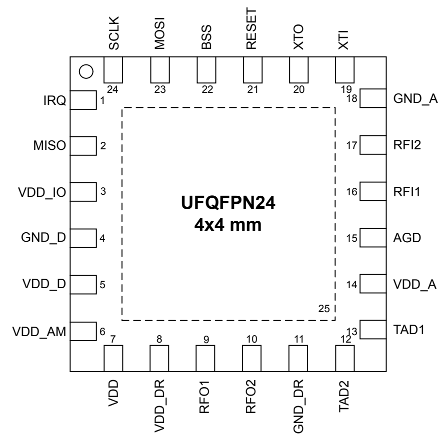
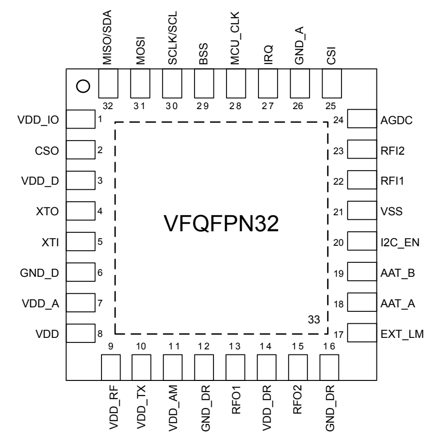
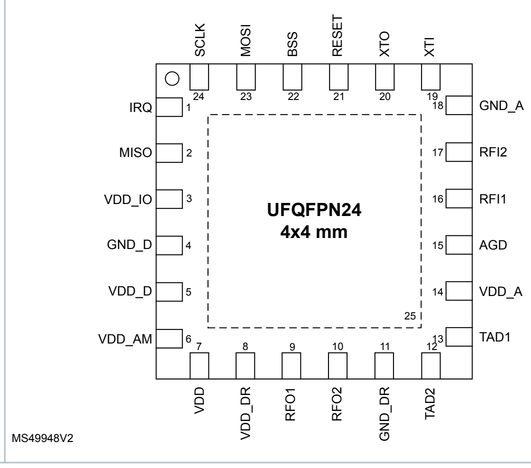
<table>
<tr><th>ST25R3911B</th><th>ST25R200</th></tr>
<tr><td>
</td><td>
</td></tr>
</table>

Table 4. Comparison between ST25R3916B and ST25R200

<table>
<tr><th>ST25R3916B</th><th>ST25R200</th></tr>
<tr><td>
</td><td>
</td></tr>
</table>

2.2 Pinout differences

The table below describes the main pinout differences when migrating from the ST25R39xx to the ST25R200:

Table 5. **Pinout differences**

Element	Description
Reset pin	The ST25R200 introduces a reset pin (active high) with the possibility to perform a reset via an external pin. This pin can be used to put the device in the lowest power consumption mode, when the NFC functionality is not needed. This pin is optional, and must be tied to GND when it is not used. The device can also be reset via direct command, as the previous devices.
Capacitive sensing pins	On ST25R200, capacitive sensing is no longer available and only inductive measurements are available. Due to that reason, the CSI/CSO pins are no longer present.
Exposed pad	On ST25R200, pin 25 (exposed pad) is not only used for thermal release, but also to connect to GND. A good connection to the PCB ground plane via multiple vias is extremely important to ensure proper operation.

2.3 Design considerations

Although most pins/signals are common between ST25R devices, it is important to consider the specifics of each device when designing boards/systems. For the ST25R200, these are detailed in the AN5984.

Similarly, different antenna matching characteristics must be considered between ST25R39xx designs and ST25R200.

Detailed information on the matching specifics, as well as guidance on the development and verification process are provided in the AN6065.

3 Feature description and comparison

This section lists and compares the main features for ST25R39xx and ST25R200.

3.1 Power management

ST25R200 features two positive supply pins: V_{DD} and V_{DD_IO}.

V_{DD} is the main power supply pin. It supplies the analog and digital blocks through two regulators (V_{DD_A}, V_{DD_D}), as well as the transmitter supplies (V_{DD_DR}, V_{DD_AM}). The supported V_{DD} supply range goes from 2.7 V to 5.5 V.

Regulated voltage can be automatically adjusted to have the maximum possible value while still having a good PSRR. All regulator pins (except V_{DD_AM}) also have corresponding negative supply pins, which are externally connected to the ground potential (V_{SS}).

The recommended blocking capacitors for supply pins are the following:

- V_{DD}, V_{DD_DR}, V_{DD_D}, V_{DD_AM} (pins 5, 6, 7, 8): 2.2 µF in parallel with 10 nF.
- AGD, V_{DD_IO} (pins 3, 15): 1 µF in parallel with 10 nF.

On ST25R200, the regulation of V_{DD_DR} can be set by a defined voltage configuration, as in ST25R39xx, or by a target voltage drop in reference to the voltage supplied at V_{DD}.

For further information, refer to the ST25R200 datasheet (DS13658).

Figure 1. System diagram (differential antenna driving)

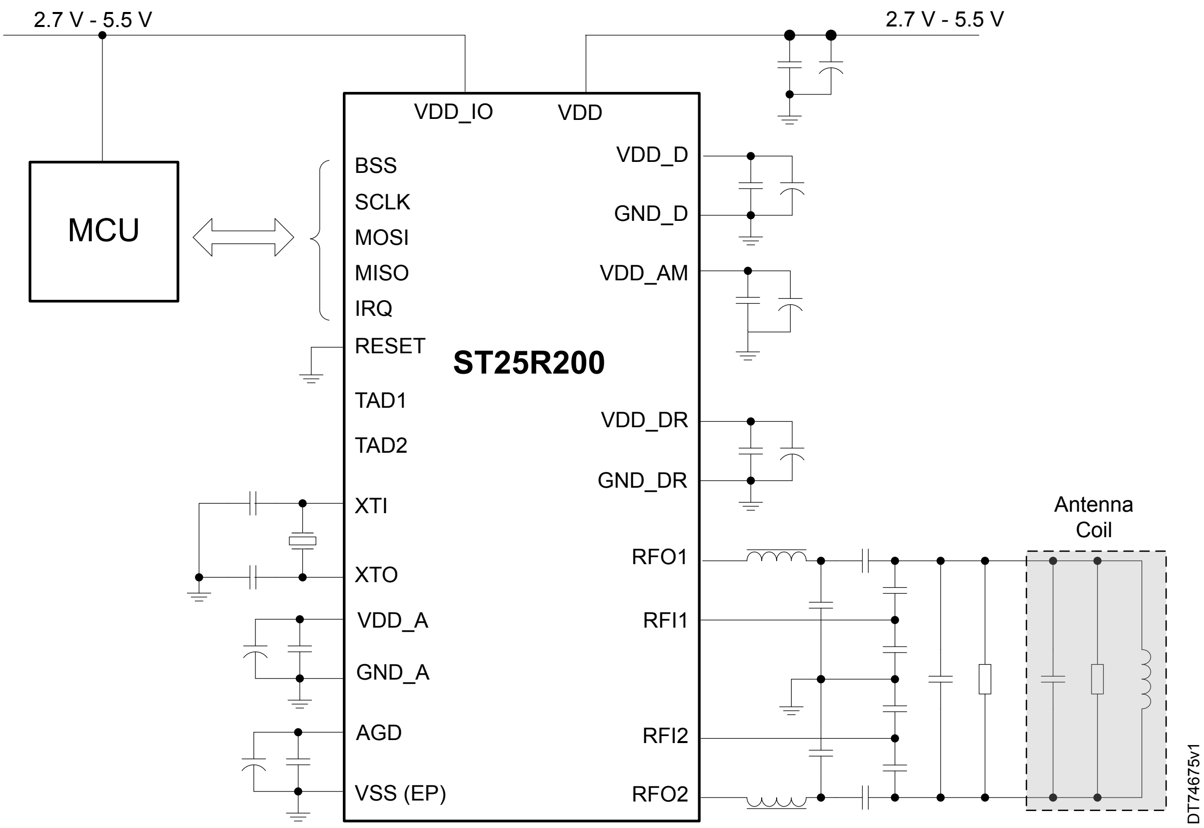
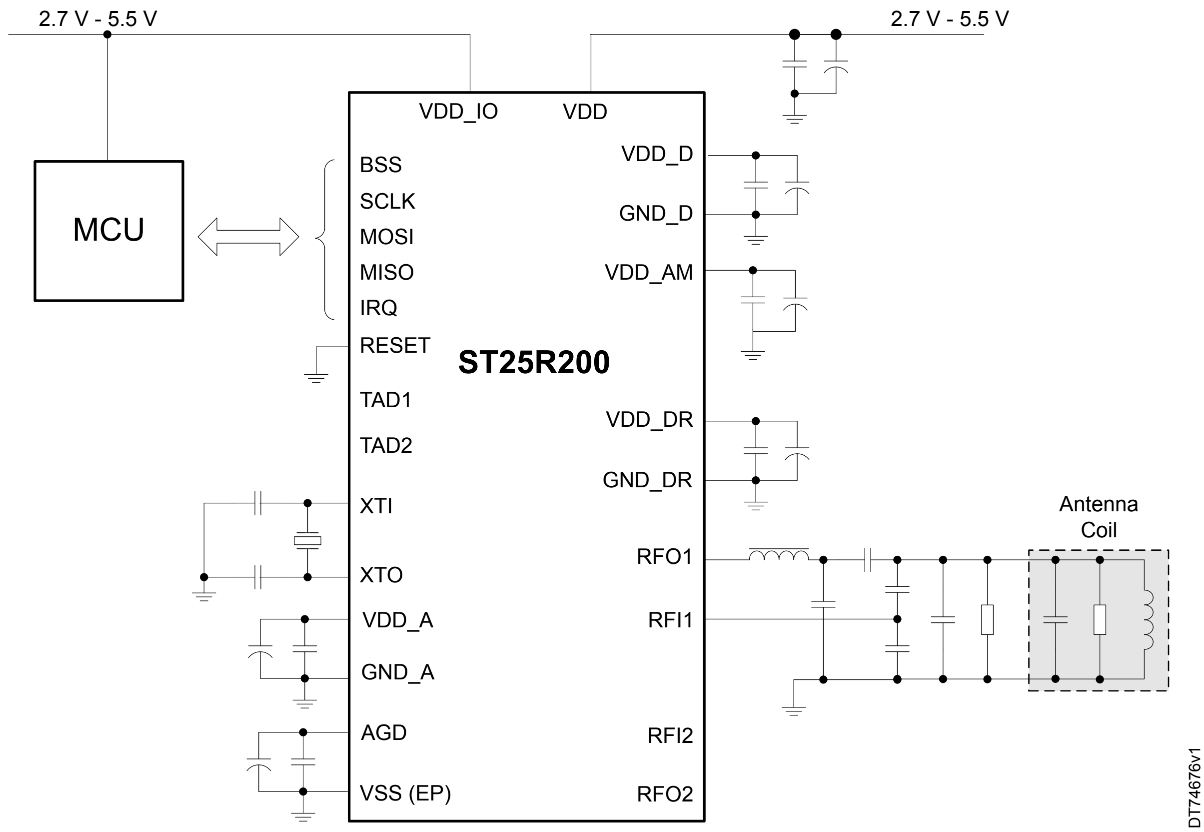


Figure 2. System diagram (single-ended driving)



3.2 Automatic antenna tuning (AAT)

Unlike the ST25R39xx, the ST25R200 cannot change the characteristics of the antenna matching and cannot retune the antenna during operation. The matching network mounted on the antenna defines the tuning.

3.3 Serial interface

The ST25R200 uses SPI as the interface to the host, supporting speeds of up to 6 Mbit/s. Any system communicating with ST25R39xx via SPI is able to directly interface with ST25R200 (maintaining SPI mode and speed).

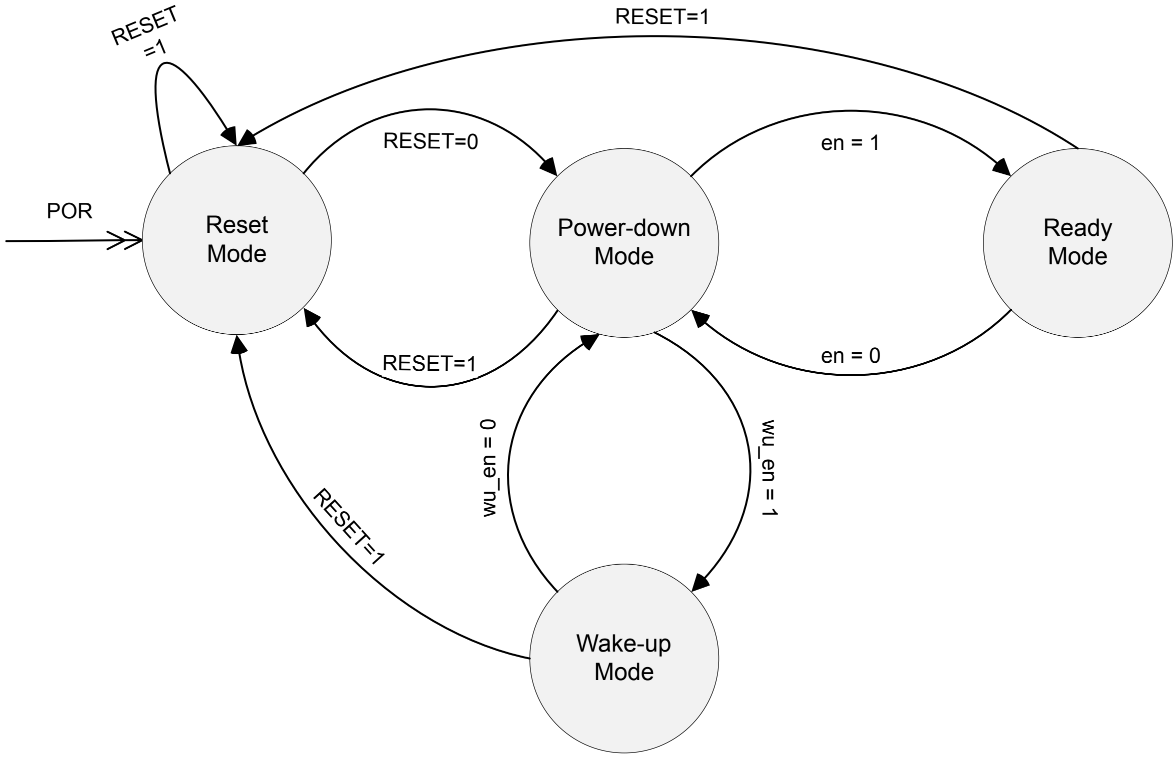
3.4 Operating modes

ST25R200 introduces a new mode, which is controlled by the RESET pin.

In reset mode, the device is disabled, and power consumption is minimal.

The power-down mode is used to enter the wake-up and ready modes.

Figure 3. Operating modes



3.5 Wake-up mode

The wake-up (WU) mode on the ST25R200 is distinct from the one on ST25R39xx. The table below highlights the differences:

Table 6. Wake-up mode comparison

Wake-up mode features	ST25R39xx	ST25R200
Capacitive sensing	Yes	No
Inductive sensing	AM/PM channels	I/Q decomposition
Calibration required	No	Yes
Configurable measurement of pulse duration	No	Yes
Configurable wake-up trigger criteria	No	Yes

3.6 External field detector

In contrast with the ST25R39xx, the ST25R200 does not have a dedicated external field detector block.

ST25R200 can sense the presence of its own or external RF field. However, the mechanisms to support the RF collision avoidance of the NFC Forum are not available.

3.7 NFC technologies and modes

The ST25R200 does not support all the existing technologies and modes foreseen by the NFC Forum.

The following modes are **not supported**:

- NFC-F
- P2P (PCM or ACM)
- Listen mode

3.8 NFC-V/ISO15693 framing

On the ST25R39xx, NFC-V is indirectly available, via the stream mode. In stream mode, the host handles encoding or decoding. The NFC-V/ISO15693 framing is fully supported and integrated.

4 Operation and interface

The ST25R200 has comparable operation modes and interfaces compared to the previous ST25Rxxx multipurpose NFC transceivers. Despite the similarities, each device presents its own specificities.

The use of the STMicroelectronics NFC library (RFAL) is recommended: RFAL supports all the required technologies and protocols while encapsulating all the existing ST25Rxxxx devices (ST25R3911, ST25R3916, ST25R95, ST25R200). It is not recommended to modify existing outdated ST25R drivers.

If RFAL is already being used on the project, the process to migrate from ST25R39xx to ST25R200 is as simple as switching the drivers to compile. Further details are provided in the following sections.

4.1 Interface scheme

Compared to the ST25R39xx, the ST25R200 uses a different addressing scheme, allowing:

- Up to 95 addressable registers,
- Up to 64 command codes.

For further details, refer to the ST25R200 datasheet (DS13658).

4.2 Registers

Note: *Despite the fact that several configuration and bit fields are similar and have analogous functionality, the register addresses on the ST25R200 have no direct mapping into previous ST25R devices.*

4.3 FIFO

The ST25R200 comprises a FIFO with 256 bytes.

The ST25R3911B and ST25R3916(B) have a FIFO of 96 and 512 bytes respectively.

FIFO water levels on the ST25R200 are fixed at 128 bytes in both directions (transmission and reception).

4.4 Interrupts

The ST25R200 has three interrupt status registers that hold the IRQ information.

Interrupt signals on the ST25R200 are identical as on the ST25R39xx, except for the following:

Table 7. Differences in interrupt signals

Interrupt signal	Description
I_subc_start	New IRQ indicating that a detected subcarrier.
I_hfe	Renamed from I_err1, indicates a soft framing error.
I_sfe	Renamed from I_err2, indicates a hard framing error.
I_wuq	New IRQ indicating wake-up Q-channel measurement
I_wui	New IRQ indicating wake-up I-channel measurement

4.5 Timers

Timer structure on the ST25R39xx and ST25R200 is similar:

- MRT
- NRT
- GPT
- WUT

However, the MRT in ST25R200 permits additional steps for increased precision.

5 Migrating existing software projects

The use of the STMicroelectronics NFC library (RF abstraction layer) is strongly recommended.

RFAL supports all the required technologies and protocols, while encapsulating all the existing ST25R devices (ST25R3911, ST25R3916, ST25R95, ST25R200).

The STMicroelectronics NFC library is periodically updated to introduce new features, wider NFC device support, and fixes. Its compliance with the latest versions of the relevant standards is continuously ensured.

Patching and tweaking of existing non-RFAL ST25R drivers it is not recommended. An incomplete driver that cannot be supported is unable to receive official updates.

5.1 RFAL projects

The RF abstraction layer (RFAL) is structured in a way that the ST25RXXX device underneath is encapsulated.

Migrating an existing ST2539xx project only requires the modification of the RFAL HAL to ST25R200:

1. On the project compilation list or makefile, replace the preprocessor instruction defining the targeted ST25R device to ST25R200.
2. Replace the included folder from `rfal/source/st25rxxx` to `rfal/source/st25r200`.
3. Switch the following modules:

Table 8. Modules

ST25R3911B	ST25R3916	ST25R200
`rfal_rfst25r3911`	`rfal_rfst25r3916`	`rfal_rfst25r200`
`st25r3911`	`st25r3916`	`st25r200`
`st25r3911_com`	`st25r3916_com`	`st25r200_com`
`st25r3911_interrupt`	`st25r3916_irq`	`st25r200_irq`
-	`st25r3916_led`	-

5.2 Non-RFAL projects

The standard procedure is to use the RFAL, as mentioned in Section 5.1: RFAL projects.

To adapt your project accordingly, refer to the existing documentation and reference designs available on www.st.com.

Revision history

Table 9. Document revision history

Date	Version	Changes
08-Aug-2023	1	Initial release.
24-Nov-2023	2	• Updated part number in all sections. • Section Introduction: – Updated reference to STM25R100 datasheet. – Added reference to UM2890. • Updated Figure 1. System diagram (differential antenna driving) and Figure 2. System diagram (single-ended driving). • Updated Section 3.3: Serial interface.
25-Mar-2024	3	Updated Section Introduction • Added reference to AN6065.
04-Apr-2024	4	Changed the document scope from ST restricted to public.
29-May-2024	5	Updates: • Replaced device ST25R100 with ST25R200. • Minor text changes.

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